# MILITARY SPECIFICATION

## BENZALKONIUM CHLORIDE SOLUTION, USP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers Benzalkonium Chloride Solution, USP (See 6.2).

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-C-186 | - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (See 6.3) |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.



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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS.

3.1 Material. Shall be Benzalkonium Chloride Solution, USP, 10% W/V, and shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.1.1 Assay. Shall contain 10% W/V benzalkonium chloride within the applicable assay limits for the solution.

3.2 Ingredients.

3.2.1 Benzalkonium chloride. The benzalkonium chloride used in the manufacture of the solution shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.2.2 In lieu of Benzalkonium Chloride, USP, a concentrate (17 percent to 50 percent) of benzalkonium chloride in aqueous solution may be used in preparing the 10% solution. The concentrate shall comply with the following:

3.2.2.1 Identification. Shall respond to identification tests A and B for Benzalkonium Chloride Solution, USP.

3.2.2.2 Foreign amines. Shall comply with the test for foreign amines in the monograph for Benzalkonium Chloride, USP.



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3.2.2.3 Water - insoluble matter. Shall comply with the test for water-insoluble matter in the monograph for Benzalkonium Chloride, USP. The concentrate shall be diluted to contain 1 in 10 of benzalkonium chloride.

3.2.2.4 Residue on ignition. Shall comply with the test for "Residue on Ignition" in the monograph for Benzalkonium Chloride, USP. The sample of concentrate shall contain between 1 and 2 grams benzalkonium chloride and shall be accurately weighed. The 2.0 percent limit shall be calculated from the benzalkonium chloride content.

3.2.2.5 Ratio of alkyl components. Shall meet the test for ratio of alkyl components in the monograph for Benzalkonium Chloride, USP. The sample of concentrate shall be accurately weighed and shall contain about 5 grams of benzalkonium chloride.

3.2.2.6 Assay. Shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the labeled amount of benzalkonium chloride. when assayed by USP method for Benzalkonium Chloride Solution, USP.

3.2.3 Other ingredients. All other ingredients entering into the finished suspension shall be of USP or NF quality, or, if not included in either of these compendia, the ingredients shall be suitable for use in this preparation.

3.3 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the solution to the date of delivery to the Government.

3.4 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the



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contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling

| For visual examination | Inspection level | AQL (percent defective) | Unit of product |
|---|---|---|---|
| Major | S-2 | 1.0 | Filled immediate container |
| Minor | II | 2.5 | Filled immediate container |

4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE II. Classification of defects

| Categories | Defects 1/ |
|---|---|
| Major | |
| 101 | Solution not free from foreign matter. |
| 102 | Solution has crystal formation. |
| 103 | Solution not homogeneous. |
| Minor | |
| 201 | Exterior of immediate container is not free from contents or foreign matter. |
| 202 | Label not free from defacing marks and tears. |

1/ Inspection is not restricted to the classified possible defects listed above.



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4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests. Tests shall be performed to determine compliance with all requirements of this document and all referenced documents.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (See 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One bottle containing 4 fl oz (118 ml), as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit packaging. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Twelve (12) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Ninety-six (96) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.



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5.3.2 Level B. Ninety-six (96) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in paragraph 5.5.3 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed.

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed.



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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3.

6.2 This specification covers the following item(s):

| National Stock Number | Item Identification |
|---|---|
| 6505-00-149-8705 | BENZALKONIUM CHLORIDE SOLUTION, USP, 10%, 4 fl oz (118 ml) |

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1588



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-B-37451 BENZALKONIUM CHLORIDE SOLUTION, ,USP

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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